

Project Check-In: NBA Player Performance & Salary Analytics

CPSC 482 – Data-Intensive Systems

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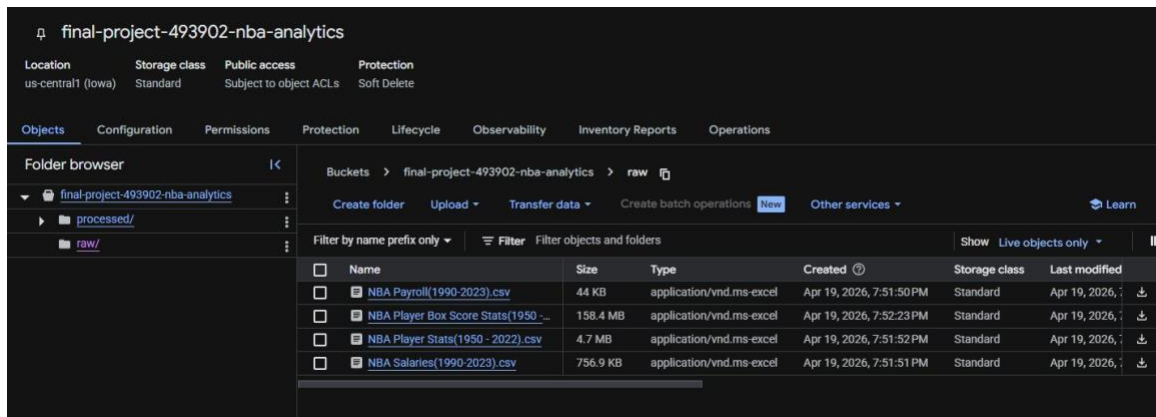
1. Progress Summary

The pipeline is approximately 65% complete. Data ingestion, cloud storage, local data processing, and BigQuery warehousing are all fully operational with real NBA data. The BigQuery ML modeling phase is underway, and the Looker Studio dashboard has not yet been built.

Pipeline Stage	Work Completed	Status
Data Ingestion	Sourced 4 CSV datasets: player box scores (1950–2022), season stats, player salaries, and team payroll	Complete
Cloud Storage	Created GCS bucket (final-project-493902-nba-analytics); uploaded all 4 raw CSVs to gs://bucket/raw/	Complete
Data Processing	Pandas transformation: cleaned column names, parsed dates, removed zero-minute rows, computed per-36 and TS%, deduplicated; output as Parquet	Complete
BigQuery Load	Loaded 4 processed Parquet tables into BigQuery dataset nba_analytics	Complete
BigQuery ML	Training data view created; salary prediction models (Linear Reg + XGBoost) ready to train	In Progress
Looker Studio	Dashboard design planned; data source connections not yet configured	Not Started

Detailed milestones:

- Data Ingestion:** Sourced four CSV datasets covering NBA player box scores (1950–2022, 1.3M rows, 158 MB), player season averages (1950–2022, 28K rows), individual player salaries (1990–2023, 15.8K rows), and team payroll (1990–2023, 966 rows). Total raw data: approximately 164 MB.
- Cloud Storage:** Created GCS bucket *final-project-493902-nba-analytics* in us-central1 (Iowa). Uploaded all four raw CSVs to the raw/ prefix and processed Parquet files to the processed/ prefix.



3. **Data Processing:** Ran a local Pandas transformation pipeline (transform_local.py) that cleaned all four datasets. Key transformations included renaming columns to consistent snake_case, parsing date strings, casting numeric types, stripping currency formatting from salary fields, filtering out zero-minute box score rows, removing duplicate entries, dropping "TOT" aggregate rows from season stats, and computing derived features including per-36-minute stats, true shooting percentage, and a home/away flag. Output: four Parquet files totaling approximately 27 MB.
4. **BigQuery Load:** Created BigQuery dataset nba_analytics and loaded all four Parquet files as native BigQuery tables. Verified row counts match the transformation output:

Table	Row Count
player_box_scores	1,183,956
player_season_stats	25,732
player_salaries	15,203
team_payroll	966

1	player_season_stats	25732
2	player_salaries	15203
3	player_box_scores	1183956
4	team_payroll	966

Field name	Type	Mode	Description	Key	Collation	Default Value	Policy Tags	Data Policies
season	INTEGER	NULLABLE						
game_id	INTEGER	NULLABLE						
player_name	STRING	NULLABLE						
team	STRING	NULLABLE						
game_date	TIMESTAMP	NULLABLE						
matchup	STRING	NULLABLE						
wins_loss	STRING	NULLABLE						
minutes	INTEGER	NULLABLE						
fg_made	INTEGER	NULLABLE						
fg_attempted	INTEGER	NULLABLE						
fg_pct	INTEGER	NULLABLE						
fg3_made	INTEGER	NULLABLE						
fg3_attempted	INTEGER	NULLABLE						
fg3_pct	INTEGER	NULLABLE						
ft_made	INTEGER	NULLABLE						
ft_attempted	INTEGER	NULLABLE						
ft_pct	INTEGER	NULLABLE						
off_rebounds	INTEGER	NULLABLE						
def_rebounds	INTEGER	NULLABLE						
rebounds	INTEGER	NULLABLE						
assists	INTEGER	NULLABLE						
steals	INTEGER	NULLABLE						
blocks	INTEGER	NULLABLE						
turnovers	INTEGER	NULLABLE						
personal_fouls	INTEGER	NULLABLE						
points	INTEGER	NULLABLE						
plus_minus	INTEGER	NULLABLE						
pts_per36	INTEGER	NULLABLE						
reb_per36	INTEGER	NULLABLE						
ast_per36	INTEGER	NULLABLE						
true_shooting_pct	INTEGER	NULLABLE						
is_home	BOOLEAN	NULLABLE						

5. **BigQuery ML (in progress):** Designed the ML training view that joins player season stats with salary data to predict inflation-adjusted salary from performance metrics (age, games played, MPG, PPG, RPG, APG, SPG, BPG, FG%, 3P%, eFG%). SQL for both a Linear Regression and XGBoost model is written and ready to execute.

2. Issues and Challenges

- **File naming inconsistencies:** The CSV filenames contained spaces and parentheses (e.g., "NBA Player Box Score Stats(1950 - 2022).csv"), which caused FileNotFoundError in the initial transformation scripts. Resolved by updating all file path references to match the exact filenames.
- **GCP project ID vs. display name:** The GCP project ID (final-project-493902) differs from the display name (final-project), which caused authentication and bucket creation errors when using the wrong identifier. Resolved by running gcloud projects list to find the correct project ID.
- **Missing historical data:** Several columns (steals, blocks, turnovers, three-point stats) are null for games before these statistics were officially tracked by the NBA. These rows are retained with null values so that basic scoring and rebounding analysis can still span the full 1950–2022 range.

3. Planned Next Steps

The remaining work focuses on completing the ML models, building the dashboard, and preparing the final deliverables:

6. **Train BigQuery ML models** (by April 28) – Execute the salary prediction SQL to train both the Linear Regression and XGBoost models. Evaluate with R^2 , MAE, and RMSE. Generate the overpaid/underpaid player predictions table.
7. **Build Looker Studio dashboard** (by May 2) – Connect Looker Studio to BigQuery views. Build the player performance explorer, league trend charts, salary efficiency analysis, and predicted vs. actual salary scatter plot.
8. **Prepare presentation** (by May 5) – Build slides covering the pipeline architecture, demo the dashboard live, and summarize key insights from the salary prediction model.
9. **Write final report and push to GitHub** (by May 9) – Document all pipeline steps with screenshots, include all scripts and SQL in the repository, and provide collaborator access.